SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS

COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 40%

Annexure A

ANALYTICAL PROBLEM SOLVING

Code of Conduct:

I **KONGARA JOGESH**(192525035) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:K.JOGESH

Annexure A

ANALYTICAL PROBLEM SOLVING

1. A university network is assigned the IP address block 192.168.20.0/24 for its internal communication. The network administrator has decided to divide this network into subnets for three different labs using fixed subnet masks. Lab A is assigned a /26 subnet, Lab B is assigned a /27 subnet, and Lab C is assigned a /28 subnet. Based on the given subnet masks, apply subnetting concepts to determine the network address of each lab. Further, identify the valid range of host IP addresses and the broadcast address for each subnet. Finally, calculate the total number of usable host IP addresses available in each lab's subnet.

2. A network administrator has configured two systems with the following details: System 1 has IP address 192.168.1.130 with subnet mask 255.255.255.192, and System 2 has IP address 192.168.1.190 with subnet mask 255.255.255.192. Using the given configuration, apply subnetting techniques to determine the subnet range, network address, and broadcast address for each system. Identify the valid host IP range for both subnets based on the given subnet mask. Calculate whether each IP address falls within its respective subnet range. Also, determine the number of usable host addresses available in each subnet.

3. An organization has been allocated the IP block 172.16.0.0/16 for its internal network. The network administrator has already decided the subnet masks for each department as follows: HR → /25, Sales → /26, IT → /27, and Admin → /28. Using the given subnet masks, apply subnetting techniques to assign suitable subnet addresses for each department. Determine the network address, valid host IP range, and broadcast address for each subnet. Based on the allocation, calculate the number of usable host addresses in each department. Finally, identify the remaining unused IP address range within the given network block.

4. A company uses the IP address 172.16.10.0/24 and has implemented subnetting using a /27 subnet mask for its internal departments. Using the given subnet mask, apply subnetting techniques to calculate the total number of subnets created and the number of usable hosts per subnet. Determine the subnet to which the IP address 172.16.10.78 belongs. Identify the network address and broadcast address for that subnet. Also, determine whether the IP address 172.16.10.95 falls within the same subnet range based on the given configuration.

5. A network uses the Internet Protocol version 6 address 2001:0db8:0000:0000:0000:ff00:0042:8329. Apply IPv6 addressing rules to compress the given address into its shortest form and then expand the

compressed address back to its full notation. Using a prefix length of /64, determine the network prefix of the given address. Also, calculate the total number of possible host addresses available within this network based on the given prefix length.

RUBRICS FOR CALCULATION

Criteria	**Weightage**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

STEP-BY-STEP SOLUTIONS

Question 1

Given Data

Base network: 192.168.20.0/24 (Class C private address, default mask 255.255.255.0)

Lab A requirement: /26 subnet mask

Lab B requirement: /27 subnet mask

Lab C requirement: /28 subnet mask

Concept Used

For any subnet mask /n, the number of host bits available is (32 - n). The block size (size of each subnet) is calculated as 2^(host bits), and the number of usable host addresses is 2^(host bits) - 2, since the first address of a block is always reserved as the Network Address and the last address is reserved as the Broadcast Address.

Lab A – /26 (Subnet mask 255.255.255.192)

Binary of mask: 11111111.11111111.11111111.11000000 → 26 network bits, 6 host bits

Block size: 2^6 = 64 → 256 - 192 = 64 (confirms same result)

Network address: 192.168.20.0

Valid host range: 192.168.20.1 to 192.168.20.62

Broadcast address: 192.168.20.63

Usable hosts: 2^6 - 2 = 62 hosts

Lab B – /27 (Subnet mask 255.255.255.224)

Binary of mask: 11111111.11111111.11111111.11100000 → 27 network bits, 5 host bits

Block size: 2^5 = 32

Network address: 192.168.20.64 (immediately after Lab A's block, which ends at .63)

Valid host range: 192.168.20.65 to 192.168.20.94

Broadcast address: 192.168.20.95

Usable hosts: 2^5 - 2 = 30 hosts

Lab C – /28 (Subnet mask 255.255.255.240)

Binary of mask: 11111111.11111111.11111111.11110000 → 28 network bits, 4 host bits

Block size: 2^4 = 16

Network address: 192.168.20.96 (immediately after Lab B's block, which ends at .95)

Valid host range: 192.168.20.97 to 192.168.20.110

Broadcast address: 192.168.20.111

Usable hosts: 2^4 - 2 = 14 hosts

Summary Table

Lab	Network Address	Host Range	Broadcast	Usable Hosts
Lab A (/26)	192.168.20.0	192.168.20.1 – .62	192.168.20.63	62
Lab B (/27)	192.168.20.64	192.168.20.65 – .94	192.168.20.95	30
Lab C (/28)	192.168.20.96	192.168.20.97 – .110	192.168.20.111	14

Interpretation

The three labs can co-exist within the same /24 block without overlapping address ranges because each subnet's block is aligned to a boundary determined by its own block size. Lab A, needing the most hosts, is given the largest block (/26), while Lab C, needing the fewest hosts, is given the smallest block (/28) — this is an example of Variable Length Subnet Masking (VLSM), which minimizes address wastage compared to using one fixed mask for all labs.

Question 2

Given Data

Subnet mask: 255.255.255.192 → /26 (11111111.11111111.11111111.11000000)

Host bits: 32 - 26 = 6 → block size = 2^6 = 64

Subnet boundaries in the 4th octet: 0–63, 64–127, 128–191, 192–255

System 1 – 192.168.1.130

Locating the block: 130 lies between 128 and 191 → belongs to the 128–191 block

Network address: 192.168.1.128

Broadcast address: 192.168.1.191

Valid host range: 192.168.1.129 to 192.168.1.190

Validity check: $129 \leq 130 \leq 190 \rightarrow$ TRUE, so 192.168.1.130 is a valid usable host address in this subnet

Usable hosts: $2^6 - 2 = 62$ hosts

System 2 – 192.168.1.190

Locating the block: 190 lies between 128 and 191 → belongs to the same 128–191 block

Network address: 192.168.1.128

Broadcast address: 192.168.1.191

Valid host range: 192.168.1.129 to 192.168.1.190

Validity check: $129 \leq 190 \leq 190 \rightarrow$ TRUE, so 192.168.1.190 is the last usable host address in this subnet

Usable hosts: $2^6 - 2 = 62$ hosts

Interpretation

Both System 1 (192.168.1.130) and System 2 (192.168.1.190) fall within the same subnet, 192.168.1.128/26, since both addresses lie between the network address (192.168.1.128) and the broadcast address (192.168.1.191). This means the two systems can communicate directly at Layer 2 / Layer 3 without needing a router, because they belong to the same broadcast domain.

Question 3

Given Data

Base network: 172.16.0.0/16 (Class B private address, 65,536 total addresses)

Department masks: HR = /25, Sales = /26, IT = /27, Admin = /28

Concept Used

Since each department needs a different number of hosts, Variable Length Subnet Masking (VLSM) is applied. Departments with larger host requirements are assigned first (largest block first), and each subsequent subnet begins immediately after the previous one ends, to avoid address overlap and wastage.

HR – /25 (Subnet mask 255.255.255.128)

Host bits / block size: $32 - 25 = 7$ host bits $\rightarrow 2^7 = 128$

Network address: 172.16.0.0

Valid host range: 172.16.0.1 to 172.16.0.126

Broadcast address: 172.16.0.127

Usable hosts: 2^7 - 2 = 126 hosts

Sales – /26 (Subnet mask 255.255.255.192)

Host bits / block size: 32 - 26 = 6 host bits → 2^6 = 64

Network address: 172.16.0.128 (next available block after HR ends at .127)

Valid host range: 172.16.0.129 to 172.16.0.190

Broadcast address: 172.16.0.191

Usable hosts: 2^6 - 2 = 62 hosts

IT – /27 (Subnet mask 255.255.255.224)

Host bits / block size: 32 - 27 = 5 host bits → 2^5 = 32

Network address: 172.16.0.192 (next available block after Sales ends at .191)

Valid host range: 172.16.0.193 to 172.16.0.222

Broadcast address: 172.16.0.223

Usable hosts: 2^5 - 2 = 30 hosts

Admin – /28 (Subnet mask 255.255.255.240)

Host bits / block size: 32 - 28 = 4 host bits → 2^4 = 16

Network address: 172.16.0.224 (next available block after IT ends at .223)

Valid host range: 172.16.0.225 to 172.16.0.238

Broadcast address: 172.16.0.239

Usable hosts: 2^4 - 2 = 14 hosts

Remaining Unused IP Address Range

Total addresses in block: 172.16.0.0/16 spans 172.16.0.0 to 172.16.255.255

Addresses used so far: 172.16.0.0 to 172.16.0.239 (by HR, Sales, IT, and Admin)

Remaining unused range: 172.16.0.240 to 172.16.255.255

Unused address count: Approximately 65,536 - 240 = 65,296 addresses remain available for future departments

Interpretation

Only a very small portion (240 addresses) of the entire /16 block has been consumed by the four departments, leaving a very large reserve (172.16.0.240 to 172.16.255.255) for future expansion, new departments, or additional VLANs, which is one of the key advantages of starting with a large address block such as a /16.

Question 4

Given Data

Base network: 172.16.10.0/24

Subnet mask applied: /27 → 255.255.255.224

Step 1: Number of Subnets

Formula: Number of subnets = 2^(borrowed bits) = 2^(new prefix - old prefix)

Borrowed bits: 27 - 24 = 3 bits borrowed from the host portion

Total subnets: 2^3 = 8 subnets

Step 2: Usable Hosts per Subnet

Host bits remaining: 32 - 27 = 5 host bits

Usable hosts: 2^5 - 2 = 30 hosts per subnet

Step 3: List of Subnet Boundaries

Block size: 256 - 224 = 32

All 8 subnets (4th octet): 0–31, 32–63, 64–95, 96–127, 128–159, 160–191, 192–223, 224–255

Step 4: Locating 172.16.10.78

Comparison: 78 lies between 64 and 95, so it falls in the 3rd subnet block

Network address: 172.16.10.64

Broadcast address: 172.16.10.95

Valid host range: 172.16.10.65 to 172.16.10.94

Conclusion: 172.16.10.78 belongs to the subnet 172.16.10.64/27 and is a valid, usable host address

Step 5: Checking 172.16.10.95

Comparison: 95 is the upper boundary of the 64–95 block

Role of .95: It is the Broadcast Address of the 172.16.10.64/27 subnet, not a host address

Conclusion: 172.16.10.95 lies within the same subnet range (172.16.10.64/27) as 172.16.10.78, but it CANNOT be assigned to a host since it is reserved for broadcast traffic

Interpretation

Both addresses, 172.16.10.78 and 172.16.10.95, belong to the same /27 subnet (172.16.10.64/27). However, only 172.16.10.78 can be configured on a host device; 172.16.10.95 is reserved by the protocol for broadcasting messages to every device in that particular subnet and must never be manually assigned to a network interface.

Question 5

Given Data

Full IPv6 address: 2001:0db8:0000:0000:0000:ff00:0042:8329

Total address length: 128 bits, divided into 8 groups of 16 bits (4 hex digits each)

Prefix length required: /64

Step 1: Compression Rules

Rule 1: Leading zeros within each 16-bit group may be omitted

Rule 1 applied: 2001:db8:0:0:0:ff00:42:8329

Rule 2: One single contiguous sequence of all-zero groups may be replaced with a double colon '::' (this shorthand can be used only once in an address)

Rule 2 applied: The three consecutive zero groups become '::'

Compressed (shortest) form: 2001:db8::ff00:42:8329

Step 2: Expansion Back to Full Notation

Procedure: Replace '::' with the exact number of all-zero groups needed to make up 8 groups in total, then pad every group back to 4 hex digits

Groups restored: 2001:db8::ff00:42:8329 has 5 explicit groups, so '::' expands to 3 zero groups (8 - 5 = 3)

Expanded (full) form: 2001:0db8:0000:0000:0000:ff00:0042:8329

Step 3: Network Prefix for /64

Rule: A /64 prefix means the first 64 bits (the first 4 groups of 16 bits each) form the network portion, and the remaining 64 bits form the interface identifier (host portion)

First 4 groups: 2001 : 0db8 : 0000 : 0000

Network prefix: 2001:0db8:0000:0000::/64, written in compressed form as 2001:db8::/64

Step 4: Total Possible Host Addresses

Host bits available: 128 - 64 = 64 bits

Formula: Total addresses = 2^(host bits)

Total possible host addresses: 2^64 addresses (unlike IPv4, no addresses are subtracted for network/broadcast, since IPv6 has no broadcast address)

Interpretation

A /64 prefix is the standard subnet size recommended for IPv6 LANs. It provides 2^64 (over 18 quintillion) unique interface addresses per subnet, which is far more than any physical network segment will ever need — this generous allocation simplifies address auto-configuration (SLAAC) and eliminates the address-exhaustion concerns that exist in IPv4.